

1) Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Step 1: Open Wireshark

- Install and open Wireshark.
- Select your active network interface, such as Wi-Fi or Ethernet.

Step 2: Start the Capture

- Start capturing packets on the selected interface.
- Open a web browser and visit a website such as instagram.com.
- Allow the page to load and generate network traffic.

Step 3: Stop and Review

- Stop the capture after enough traffic has been collected.
- Review the Protocol column in the packet list.

Protocols You May Observe

- DNS — used for domain-name resolution.
- TCP — transport protocol used by many application connections.
- TLS — used to secure HTTPS traffic.
- HTTP/HTTP2 or QUIC — may appear depending on the website and connection.
- ARP — used for local network address resolution.

Packet Record

- Protocol 1: _____
- Protocol 2: _____
- Protocol 3: _____
- Protocol 4: _____
- Protocol 5: _____

2) Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents.

Step 1: Select Three Packets

- Choose three packets from the capture.
- Use the Protocol column and Info details to help identify what each packet represents.

Step 2: Map the Packets to OSI Layers

- HTTP or another application protocol → mainly Application layer.
- TCP → mainly Transport layer.
- IP → mainly Network layer.
- Ethernet/Wi-Fi → mainly Data Link layer.
- The exact mapping depends on the packet selected and the protocol information visible in Wireshark.

Packet Analysis Table

- Packet 1 — Protocol: _____ — OSI Layer: _____ — Evidence/Info: _____.
- Packet 2 — Protocol: _____ — OSI Layer: _____ — Evidence/Info: _____.
- Packet 3 — Protocol: _____ — OSI Layer: _____ — Evidence/Info: _____.

3) Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily.

OSI Model Diagram

OSI Model - 7 Layers with Examples

7. Application	Web browsing, WhatsApp
6. Presentation	Data format, encryption
5. Session	Session establishment and management
4. Transport	TCP / UDP, ports
3. Network	IP addressing and routing
2. Data Link	Ethernet / Wi-Fi frames, MAC
1. Physical	Cables, radio signals, bits

- The reference diagram below shows the seven OSI layers and simple examples. You can recreate it in your preferred drawing tool.

Layer Examples

- Application Layer — Web browsing or WhatsApp application activity.
- Presentation Layer — Data formatting, encoding, and encryption used when application data is prepared or protected.
- Session Layer — Establishing and managing a communication session.
- Transport Layer — TCP/UDP communication and port numbers.
- Network Layer — IP addressing and routing.
- Data Link Layer — Ethernet/Wi-Fi frames and MAC addresses.
- Physical Layer — Cables, radio signals, and transmission of bits.

4) Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Step-by-Step Example

- Application Layer — You type and send the WhatsApp message. The application prepares the message for communication.
- Presentation Layer — The message data can be formatted, encoded, and protected/encrypted before transmission.
- Transport Layer — Transport protocols handle delivery between endpoints; ports and transport information are used to carry application data.
- Network Layer — IP addressing helps routers move the packet toward the destination network.
- Data Link Layer — The network packet is placed into a local Ethernet or Wi-Fi frame for transmission.
- Physical Layer — The frame is ultimately transmitted as electrical, radio, or optical signals.

At the Receiver

- The receiving device processes the data through the layers in reverse order, removing the relevant headers/encapsulation information and delivering the message to the application.

Conclusion

- Sending a WhatsApp message involves multiple OSI layers working together. Each layer performs a different function, from application data preparation to transmission across the physical network.